2025-2026 M.Sc. in Data Science and Analytics

Predicting and Explaining Traffic Collision Severity in Toronto
Using Explainable Machine Learning

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Objective

Traffic collisions remain a critical public safety concern in Toronto, causing hundreds of fatalities and thousands of injuries every year. Despite over 800,000 historical collision records being publicly available, most analyses remain descriptive and lack predictive capability. This research develops an **explainable machine learning framework** to predict traffic collision severity — classified as **Minor** (property damage only), **Major** (one or more injuries), or **Fatal** (one or more fatalities) — using **677,056** cleaned collision records (2014–2026) merged with hourly weather data from the Government of Canada's Toronto City weather station (Station ID 31688). Five classifiers (Logistic Regression, Random Forest, XGBoost, Gradient Boosting, Neural Network) are trained and compared. Five balancing strategies address the extreme class imbalance (Fatal = 0.1%). SHAP (SHapley Additive Explanations) provides global and local model interpretations. DBSCAN spatial clustering identifies **36 fatal collision hotspot zones** across Toronto to support the city's **Vision Zero** road safety strategy.

Background

Data Sources

- City of Toronto Open Data: 809,034 collision records (2014–2026) — date, time, GPS, neighbourhood, involvement flags, injuries, fatalities
- Gov. of Canada Climate Data: hourly weather Station ID 31688 — temperature, humidity, dew point, precipitation (wind speed/visibility: 100% null, excluded)
- 100% merge match on date-hour key. 131,978 invalid GPS records removed → 677,056 valid records with 37 engineered features

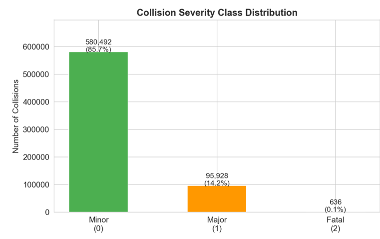


Figure 1: Severity Class Distribution

Severity Class Distribution

Severity Class	Count	%
Minor (0) — Property damage	580,492	85.7%
Major (1) — Injury	95,928	14.2%
Fatal (2) — Fatality	636	0.1%
Total	**677,056**	**100%**

Key EDA Findings

- **Hour of Day:** Minor peaks at 7–9 AM & 3–6 PM (~49,000/hr); late-night hours show proportionally higher fatal outcomes
- **Day of Week:** Friday highest (~112,000); Sunday lowest (~70,000) — commuter traffic and end-of-week fatigue
- **Seasonal:** January most collisions (~62,000); April fewest (~47,000) — winter road conditions
- **Weather:** Major proportion decreases under adverse conditions 14.2%→13.3% — risk compensation effect
- **Neighbourhood:** Wexford/Maryvale (18), West Humber-Clairville (17), South Parkdale (14) lead fatal locations
- **Correlation:** All features show <0.03 direct linear correlation with severity — non-linear models required

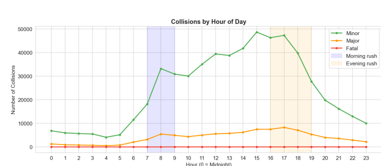


Figure 2: Collisions by Hour of Day

Results

Exp 1 — Model Performance (n=101,559)

Figure 3: Model Performance Comparison

Model	Acc.	F1-W	F1-M	AUC
XGBoost ★	**0.9003**	**0.8917**	**0.5185**	**0.7839**
Rnd. Forest	0.8926	0.8862	0.5141	0.7809
Grad. Boost	0.8922	0.8848	0.5071	0.7823
Neural Net	0.8823	0.8791	0.4992	0.7624
Log. Regr.	0.8696	0.8692	0.4827	0.7452

Exp 2 — Balancing Strategy Comparison

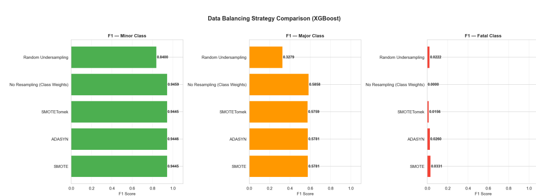


Figure 4: F1 Scores per Class by Strategy

Exp 3 — Threshold Tuning (XGBoost)

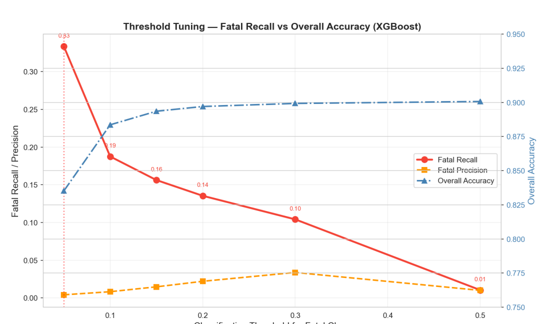


Figure 5: Threshold 0.05 → Fatal recall 4.2%→33.3% (8× gain)

SHAP — Global Feature Importance (All Classes)

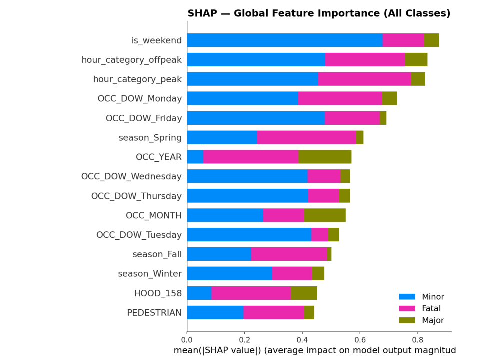


Figure 6: SHAP Global — is_weekend & hour_category dominate

SHAP — Top 10 Fatal Risk Factors

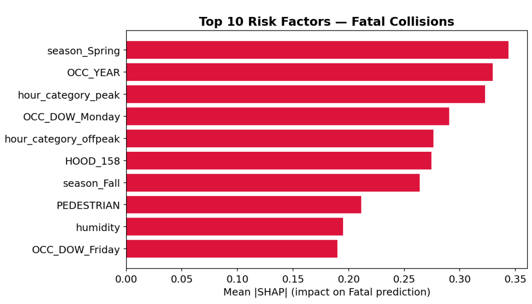


Figure 7: season_Spring (0.34), OCC_YEAR (0.33), hour_peak (0.32)

Toronto Fatal Collision Hotspot Map

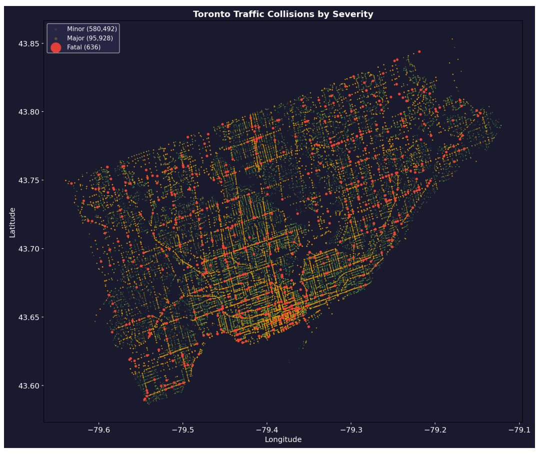


Figure 8: 36 DBSCAN hotspot zones — road network visible from density

Methodology

Five-Phase Research Framework

- **Phase 1:** Load 809,034 collisions + hourly weather; merge on date-hour (100% match)
- **Phase 2:** Remove 131,978 invalid GPS rows; fill NaN; 3-class target; engineer 8 features (season, hour_category, weather flags, pedestrian/cyclist, neighbourhood_freq)
- **Phase 3:** EDA — 8 visualizations: temporal, spatial, weather impact, correlation heatmap
- **Phase 4:** Stratified 70/15/15 split; SMOTE training only (406,344/class, 1.2M total); 5 classifiers on held-out test (n=101,559)
- **Phase 5:** SHAP TreeExplainer (2,000 records) — global, top-10, waterfall; DBSCAN (eps=300m, min=3) on 636 fatal GPS

Balancing Strategies

- No Resampling (Class Weights)
- **SMOTE** — primary strategy selected
- ADASYN — adaptive boundary sampling
- SMOTETomek — SMOTE + noise cleaning
- Random Undersampling — reduce majority

Split & Evaluation

Train 70% (473,939) | Val 15% (101,558) | Test 15% (101,559)

Metrics: F1 Macro (primary) | Accuracy | F1 Weighted | ROC-AUC

Tools: Python | Scikit-learn | XGBoost | imbalanced-learn | SHAP | Folium | DBSCAN

Figure 9: 5-Fold Cross-Validation (XGBoost)

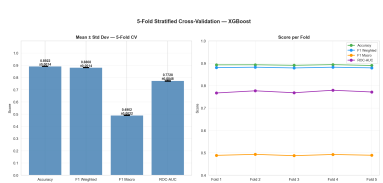
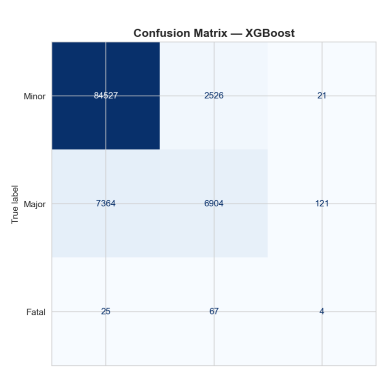


Figure 9: Accuracy 0.892 ±0.001 across 5 folds — stable, no overfitting

Figure 10: XGBoost Confusion Matrix



Minor 97.1% | Major 48.0% | Fatal 4.2% recall at default threshold

Conclusions

- **XGBoost + SMOTE is the best configuration:** Accuracy 90.0%, F1 Macro 0.519, ROC-AUC 0.784, outperforming all four classifiers. 5-fold CV confirms stability (Accuracy 0.892 ± 0.001) — no overfitting.
- **Threshold 0.05 recommended for deployment:** Increases Fatal recall from 4.2% to 33.3% (8× improvement) without retraining. Missing a fatal collision is far costlier than a false alarm.
- **No single balancing strategy dominates:** SMOTE maximizes overall F1 Macro (0.519); Random Undersampling maximizes Fatal recall (72.9%). Optimal choice depends on deployment context.
- **SHAP provides actionable insights:** Top fatal predictors: season_Spring (0.34), OCC_YEAR (0.33), hour_category_peak (0.32), HOOD_158 (0.27), PEDESTRIAN (0.21). Enables targeted seasonal campaigns and spatial interventions.
- **36 DBSCAN hotspot zones** across Toronto — top priority: Wexford/Maryvale (18), West Humber-Clairville (17), South Parkdale (14) for safety cameras, protected crossings, speed reductions.
- **Limitations:** Wind speed & visibility unavailable (100% null at Station 31688); single station limits spatial weather coverage; Fatal class has only 636 real records, constraining minority learning.

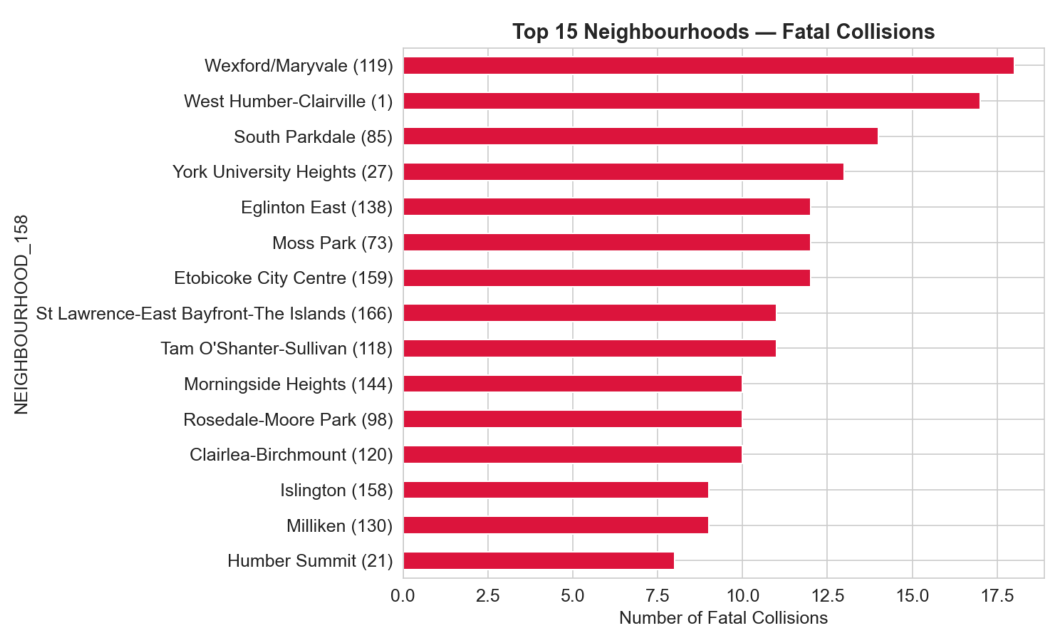


Figure 11: Top 15 Neighbourhoods by Fatal Collision Count